On the flip side, prefix-tuning can be used to understand what "skills" a model has: if prefix-tuning for a task fails, then the model likely lacks one of the key "skills" for that task.

Limitations. The present analysis is largely limited to prefixing with prompts, soft prompts and for prefix-tuning. While our theoretical results hold for suffix-tuning, they do not necessarily apply to suffixing with prompts or soft prompts. That is because the deeper representations for prompt and soft prompt suffixes would depend on the previous positions. This does not apply to suffix-tuning as it fixes all intermediate representations. Therefore, whether suffixing is more expressive than prefixing remains an open question. Separately, while we provided evidence towards context-based fine-tuning methods being parameter inefficient learners, the formal analysis of the conditions under which they may be universal approximators remain an open question. Finally, we mostly considered simple toy problems. In practice, however, language models are pretrained with very large datasets and can pick up very complex behaviors. Hence, the extent to which the limitations we demonstrated apply to large-scale pretrained transformers also remains for future work.

8 Conclusion

This paper formally showed that fine-tuning techniques working in embedding space, such as soft prompting and prefix-tuning, are strictly more expressive than prompting which operates in the discrete token space. However, we then demonstrated that despite this larger expressivity, prefix-tuning suffers from structural limitations that prevent it from learning new attention patterns. As a result, it can only bias the output of the attention layer in a direction from a subspace of rank equal to the size of the prefix. We showed that this results in practical limitations by constructing minimal transformers where prefix tuning fails to solve a simple task. This result seems to be at odds with the empirical success of prefix-tuning. We provided explanations towards that. First, we showed that prefix-tuning can easily elicit a skill the pretrained model already has and can even learn a new task, if it has picked up the skills to solve it during pretraining. Second, we showed that the effect of the prefix-induced bias is more complicated and powerful when combined with downstream non-linear operations. However, it appears to be still less expressive than full fine-tuning.

Reproducibility statement

In order to facilitate the reproduction of our empirical results, validating our theoretical results, and further studying the properties of context-based fine-tuning, we release all our code and resources used in this work. Furthermore, in Appendix A we offer explicit constructions of transformers with the properties discussed in Section 3. We also provide Python implementations of these constructions that validate their correctness.

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References

Ekin Akyürek, Dale Schuurmans, Jacob Andreas, Tengyu Ma, and Denny Zhou. 2022. What learning algorithm is in-context learning? Investigations with linear models. In *International Conference on Learning Representations*.

Jiaqi Bai, Zhao Yan, Jian Yang, Xinnian Liang, Hongcheng Guo, and Zhoujun Li. 2023. Knowprefix-tuning: A two-stage prefix-tuning framework for knowledge-grounded dialogue generation. *arXiv preprint arXiv:2306.15430*.

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *Advances in Neural Information Processing Systems*.

YunSeok Choi and Jee-Hyong Lee. 2023. CodePrompt: Task-agnostic prefix tuning for program and language generation. In *Findings of the Association for Computational Linguistics: ACL 2023*.

Ana Santos Costa, Montserrat Comesaña, and Ana Paula Soares. 2022. PHOR-in-One: A multilingual lexical database with PHonological, ORthographic and PHonographic word similarity estimates in four languages. *Behavior Research Methods*.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*.

Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. 2021. A mathematical framework for transformer circuits. *Transformer Circuits Thread*.

Karen Hambardzumyan, Hrant Khachatrian, and Jonathan May. 2021. WARP: Word-level Adversarial ReProgramming. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*.

Mohamad H Hassoun. 1995. *Fundamentals of artificial neural networks*. MIT press.

Junxian He, Chunting Zhou, Xuezhe Ma, Taylor Berg-Kirkpatrick, and Graham Neubig. 2021a. Towards a unified view of parameter-efficient transfer learning. In *International Conference on Learning Representations*.

Tianxing He, Jun Liu, Kyunghyun Cho, Myle Ott, Bing Liu, James Glass, and Fuchun Peng. 2021b. Analyzing the forgetting problem in pretrain-finetuning of open-domain dialogue response models. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*.

Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin De Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. 2019. Parameter-efficient transfer learning for NLP. In *International Conference on Machine Learning*.

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.

Zhiqiang Hu, Yihuai Lan, Lei Wang, Wanyu Xu, Ee-Peng Lim, Roy Ka-Wei Lee, Lidong Bing, and Soujanya Poria. 2023. LLM-Adapters: An adapter family for parameter-efficient fine-tuning of large language models. *arXiv preprint arXiv:2304.01933*.

Sarthak Jain and Byron C. Wallace. 2019. Attention is not explanation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. 2020. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*.

Andrej Karpathy. 2020. minGPT GitHub Repository.

Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. Large language models are zero-shot reasoners. *Advances in Neural Information Processing Systems*.

Jannik Kossen, Tom Rainforth, and Yarin Gal. 2023. In-context learning in large language models learns label relationships but is not conventional learning. *arXiv preprint arXiv:2307.12375*.

Emanuele La Malfa, Aleksandar Petrov, Christoph Weinhuber, Simon Frieder, Ryan Burnell, Anthony G. Cohn, Nigel Shadbolt, and Michael Wooldridge. 2023. The ARRT of Language-Models-as-a-Service: Overview of a new paradigm and its challenges. *arXiv preprint arXiv:2309.16573*.

Brian Lester, Rami Al-Rfou, and Noah Constant. 2021. The power of scale for parameter-efficient prompt tuning. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*.

Xiang Lisa Li and Percy Liang. 2021. Prefix-Tuning: Optimizing continuous prompts for generation. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*.

Vladislav Lialin, Vijeta Deshpande, and Anna Rumshisky. 2023. Scaling down to scale up: A guide to parameter-efficient fine-tuning. *arXiv preprint arXiv:2303.15647*.

Valerii Likhosherstov, Krzysztof Choromanski, and Adrian Weller. 2021. On the expressive power of self-attention matrices. *arXiv preprint arXiv:2106.03764*.

Pengfei Liu, Weizhe Yuan, Jinlan Fu, Zhengbao Jiang, Hiroaki Hayashi, and Graham Neubig. 2023. Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing. *ACM Computing Surveys*.

Xiao Liu, Kaixuan Ji, Yicheng Fu, Weng Tam, Zhengxiao Du, Zhilin Yang, and Jie Tang. 2022. P-Tuning: Prompt tuning can be comparable to fine-tuning across scales and tasks. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*.

Yun Luo, Zhen Yang, Fandong Meng, Yafu Li, Jie Zhou, and Yue Zhang. 2023. An empirical study of catastrophic forgetting in large language models during continual fine-tuning. *arXiv preprint arXiv:2308.08747*.

Bonan Min, Hayley Ross, Elior Sulem, Amir Pouran Ben Veyseh, Thien Huu Nguyen, Oscar Sainz, Eneko Agirre, Ilana Heintz, and Dan Roth. 2021. Recent advances in natural language processing via large pre-trained language models: A survey. *ACM Computing Surveys*.

Jishnu Mukhoti, Yarin Gal, Philip H. S. Torr, and Puneet K. Dokania. 2023. Fine-tuning can cripple your foundation model; preserving features may be the solution. *arXiv preprint arXiv:2308.13320*.

Linyong Nan, Dragomir Radev, Rui Zhang, Amrit Rau, Abhinand Sivaprasad, Chiachun Hsieh, Xiangru Tang, Aadit Vyas, Neha Verma, Pranav Krishna, Yangxiaokang Liu, Nadia Irwanto, Jessica Pan, Faiaz Rahman, Ahmad Zaidi, Mutethia Mutuma, Yasin Tarabar, Ankit Gupta, Tao Yu, Yi Chern Tan, Xi Victoria Lin, Caiming Xiong, Richard Socher, and Nazneen Fatema Rajani. 2021. DART: Open-domain structured data record to text generation. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*.

Yawen Ouyang, Yongchang Cao, Yuan Gao, Zhen Wu, Jianbing Zhang, and Xinyu Dai. 2023. On prefix-tuning for lightweight out-of-distribution detection. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*.

Abhishek Panigrahi, Nikunj Saunshi, Haoyu Zhao, and Sanjeev Arora. 2023. Task-specific skill localization in fine-tuned language models. In *International Conference on Machine Learning*.

Aleksandar Petrov, Philip HS Torr, and Adel Bibi. 2024. Prompting a pretrained transformer can be a universal approximator. *arXiv preprint arXiv:2402.14753*.

Guanghui Qin and Jason Eisner. 2021. Learning how to ask: Querying LMs with mixtures of soft prompts. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*.